

```

mysql> -- Create Departments Table
mysql> CREATE TABLE departments (
  ->   dept_id INT PRIMARY KEY,
  ->   dept_name VARCHAR(50),
  ->   location VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> -- Insert Departments
mysql> INSERT INTO departments VALUES
  -> (1, 'CSE', 'Block A'),
  -> (2, 'IT', 'Block B'),
  -> (3, 'AI', 'Block C'),
  -> (4, 'DS', 'Block D'),
  -> (5, 'HR', 'Block E');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql>
mysql> -- Show Departments Table
mysql> SELECT * FROM departments;
+-----+-----+-----+
| dept_id | dept_name | location |
+-----+-----+-----+
| 1       | CSE       | Block A  |
| 2       | IT        | Block B  |
| 3       | AI        | Block C  |
| 4       | DS        | Block D  |
| 5       | HR        | Block E  |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> -- Create Employees Table
mysql> CREATE TABLE employees (
  ->   emp_id INT PRIMARY KEY,
  ->   name VARCHAR(50),
  ->   dept_id INT,
  ->   salary DECIMAL(10,2),
  ->   FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> -- Insert Employees
mysql> INSERT INTO employees VALUES
  -> (101, 'Alice', 1, 60000),
  -> (102, 'Bob', 2, 45000),
  -> (103, 'Charlie', 3, 70000),
  -> (104, 'David', 1, 50000),
  -> (105, 'Eva', 4, 65000),
  -> (106, 'Frank', NULL, 40000); -- Employee without department
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql>
mysql> -- Show Employees Table
mysql> SELECT * FROM employees;
+-----+-----+-----+-----+
| emp_id | name   | dept_id | salary |
+-----+-----+-----+-----+
| 101    | Alice  | 1       | 60000.00 |
| 102    | Bob    | 2       | 45000.00 |
| 103    | Charlie | 3       | 70000.00 |
| 104    | David  | 1       | 50000.00 |
| 105    | Eva    | 4       | 65000.00 |
| 106    | Frank  | NULL    | 40000.00 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
mysql> -- Create Projects Table
mysql> CREATE TABLE projects (
  ->   project_id INT PRIMARY KEY,
  ->   emp_id INT,
  ->   project_name VARCHAR(50),
  ->   FOREIGN KEY (emp_id) REFERENCES employees(emp_id)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> -- Insert Projects (valid emp_ids only)
mysql> INSERT INTO projects VALUES
  -> (201, 101, 'Library System'),
  -> (202, 102, 'Payroll System'),
  -> (203, 104, 'Attendance App'),
  -> (204, 105, 'Data Warehouse');
Query OK, 4 rows affected (0.00 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> -- Show Projects Table
mysql> SELECT * FROM projects;
+-----+-----+-----+
| project_id | emp_id | project_name |
+-----+-----+-----+
| 201        | 101    | Library System |
| 202        | 102    | Payroll System |
| 203        | 104    | Attendance App |
| 204        | 105    | Data Warehouse |
+-----+-----+-----+
4 rows in set (0.00 sec)

```

```
mysql>
mysql> -- 1. INNER JOIN (Employees with Departments)
mysql> SELECT e.emp_id, e.name, d.dept_name, e.salary
      -> FROM employees e
      -> INNER JOIN departments d ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name	salary
101	Alice	CSE	60000.00
104	David	CSE	50000.00
102	Bob	IT	45000.00
103	Charlie	AI	70000.00
105	Eva	DS	65000.00

5 rows in set (0.00 sec)

```
mysql>
mysql> -- 2. LEFT JOIN (All Employees, with Dept if available)
mysql> SELECT e.emp_id, e.name, d.dept_name, e.salary
      -> FROM employees e
      -> LEFT JOIN departments d ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name	salary
101	Alice	CSE	60000.00
102	Bob	IT	45000.00
103	Charlie	AI	70000.00
104	David	CSE	50000.00
105	Eva	DS	65000.00
106	Frank	NULL	40000.00

6 rows in set (0.00 sec)

```
mysql>
mysql> -- 3. RIGHT JOIN (All Departments, with Employees if available)
mysql> SELECT e.emp_id, e.name, d.dept_name
      -> FROM employees e
      -> RIGHT JOIN departments d ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name
101	Alice	CSE
104	David	CSE
102	Bob	IT
103	Charlie	AI
105	Eva	DS
NULL	NULL	HR

6 rows in set (0.00 sec)

```
mysql>
mysql> -- 4. FULL OUTER JOIN (Simulated with UNION in MySQL)
mysql> SELECT e.emp_id, e.name, d.dept_name
      -> FROM employees e
      -> LEFT JOIN departments d ON e.dept_id = d.dept_id
      -> UNION
      -> SELECT e.emp_id, e.name, d.dept_name
      -> FROM employees e
      -> RIGHT JOIN departments d ON e.dept_id = d.dept_id;
```

emp_id	name	dept_name
101	Alice	CSE
102	Bob	IT
103	Charlie	AI
104	David	CSE
105	Eva	DS
106	Frank	NULL
NULL	NULL	HR

7 rows in set (0.00 sec)

```
mysql>
mysql> -- 5. MULTI-TABLE JOIN (Employees with Projects and Departments)
mysql> SELECT e.name, p.project_name, d.dept_name
      -> FROM employees e
      -> INNER JOIN projects p ON e.emp_id = p.emp_id
      -> INNER JOIN departments d ON e.dept_id = d.dept_id;
```

name	project_name	dept_name
Alice	Library System	CSE
Bob	Payroll System	IT
David	Attendance App	CSE
Eva	Data Warehouse	DS

4 rows in set (0.00 sec)

```
mysql>
mysql> -- 6. CROSS JOIN (All combinations of Employees x Departments)
mysql> SELECT e.name, d.dept_name
      -> FROM employees e
      -> CROSS JOIN departments d;
```

name	dept_name
Alice	HR
Alice	DS
Alice	AI
Alice	IT
Alice	CSE
Bob	HR
Bob	DS
Bob	AI
Bob	IT
Bob	CSE
Charlie	HR
Charlie	DS
Charlie	AI
Charlie	IT
Charlie	CSE
David	HR
David	DS
David	AI
David	IT
David	CSE
Eva	HR
Eva	DS
Eva	AI
Eva	IT
Eva	CSE
Frank	HR
Frank	DS
Frank	AI
Frank	IT
Frank	CSE

30 rows in set (0.00 sec)